

Introduction to Microeconomics

Lecture 2—The Economic Method (continued): Cost of Opportunity

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- You are about to buy a 10,000 KZT alarm clock at the Technodom in Mega Silkway, in front of the NU campus. However, one of your friends tells you that the same alarm clock is on sale at the Republic Avenue Technodom, on the right bank, for 5000 KZT.
- You are about to buy a 500,000 KZT laptop at the TEchnodom in Mega Silkway. However, one of your friends tells you that the same laptop is on sale at the Republic Avenue Technodom, for 490,000 KZT.

What do you do?

The framing of the choice sometimes matters...

The Coronavirus will claim 600 lives if we do nothing. We must choose between these two programs:

- Program A: Save 200 lives with certainty.
- Program B: Save 600 lives with probability $1/3$; save zero lives with probability $2/3$.

Same Coronavirus, now consider

- Program C: 400 people will die with certainty.
- Program D: $1/3$ chance that no one will die; $2/3$ chance that all 600 will die.

Losing a 20,000 KZT bill

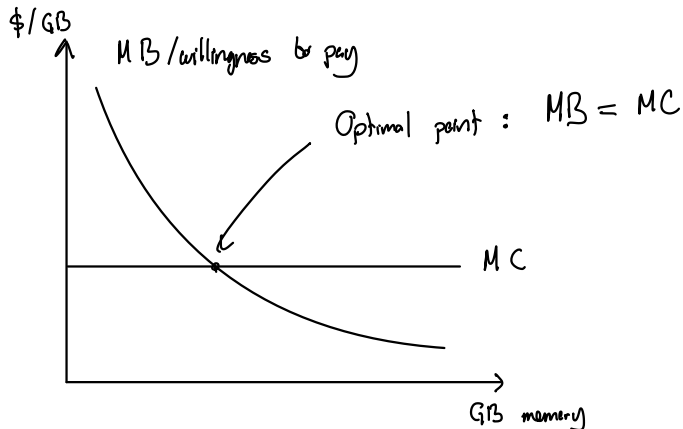
You have just arrived at the theater to buy your ticket and discover that you have lost a 20,000 KZT bill from your wallet. Do you still buy the ticket and see the play anyway?

A lost theater ticket

You have just arrived at the theater and discover that you have lost the 20,000 KZT ticket that you purchased through Ticketon earlier that day. Do you buy another ticket and see the play anyway?

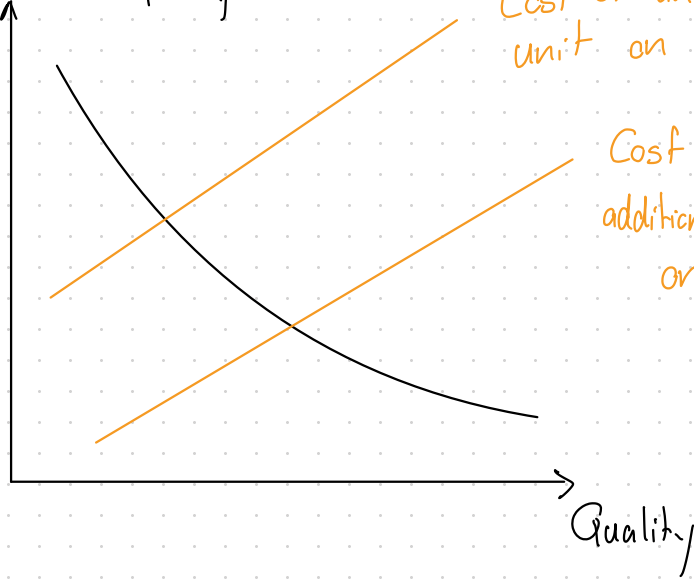
How much memory should your computer have.

Suppose that RAM can be added to your computer at \$15 per gigabyte. Suppose also that the value as measured in terms of your willingness to pay, of an additional gigabyte of memory, is as shown on the graph below.



Why airline food is so bad?

\$/unit of quality



Definition: Opportunity Cost

The opportunity cost of activity X is the total surplus (value) of the next-best alternative Y that you did not take.

Which concert should you go?

You won a free ticket to see BTS. The ticket has no resale value.

BlackPink is performing on the same night and is your next-best alternative.

Tickets to see BlackPink cost \$40.

You would be willing to pay up to \$50 to see BlackPink.

Assume there are no other costs of seeing either group. What is the opportunity cost of attending the BTS concert?

1 0

2 10

3 40

4 50

$$\begin{array}{rcl} \$50 & - & \$40 \\ B(BP) & - & C(BP) \end{array} = \$10$$

Variation 1

Now suppose that you are willing to pay as much as \$70 to see BlackPink.

The cost of a BlackPink ticket is \$30.

What is the opportunity cost of seeing BTS? $\Rightarrow \$70 - \$30 = \$40$

Variation 2

Same setup as before, but now you must buy the BTS ticket, which costs \$60. You are willing to pay as much as \$70 to see BlackPink.

The cost of a BlackPink ticket is \$30.

What is the opportunity cost of seeing BTS? $\$40 + \$60 \text{ (cost of ticket)} = \100

Variation 3

Same setup as variation 2, but now we ask, what is the minimum amount you'd have to be willing to see BTS to make their concert a better choice? *more than \$100 to make BTS a better choice.*

Variation 4

Same as before, except now you have bought a **nonrefundable** \$60 ticket to see BTS. This time, you'd have to be willing to pay at least \$40 to see BTS to make their concert a better choice $\$40$